Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

FACULTY OF SCIENCE, TECHNOLOGY, ENGINEERING AND MATHEMATICS

SCHOOL OF ENGINEERING

Electronic and Electrical Engineering

Engineering

Semester 1, 2025

Senior Sophister/M.A.I./M.Sc.

Deep Learning and its Applications

Mid-Term Exam

03/11/2025

Goldsmith Hall

Dr François Pitié

Instructions to candidates:

Answer all questions.
Key the answers in the answer sheet and only return the answer sheet.

Materials Permitted for this Examination:

Calculators

Additional information:

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Q.1. Which of the following statements are correct? (Select all correct statements)

(a) In *classification* we predict categorical outcomes.

(b) In *classification* the inputs are categorical.

(c) Classification is a case of *Supervised Learning*.

(d) Regression is a case of *Supervised Learning*.

[5.0 marks]

Q.2. Which of the following statements are correct? (mark all the statements that are true, up to +/- 1 year, so as to account for discrepensies between acceptance and publications dates).

(a) Backpropagation was published in 1986

(b) LeNet-5, by LeCun, was published in 1994

(c) AlexNet, by Alex Krizhevsky et al., was published in 2012

(d) VGG was published in 2016

[5.0 marks]

Q.3. Which of the following models with input x_1, x_2, parameters w_1, w_2 and noise $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, are linear in the parameters and can be used for Least Squares straight as? (mark all models that are linear)

(a) $y = w_1 x_1 + x_2 \varepsilon$

(b) $y = x_1 w_1 w_2 + \varepsilon$

(c) $y = w_1 + w_2(x_1 + \varepsilon)$

(d) $y = w_1 \sin(x_1 + x_2) + \varepsilon$

(e) $y = w_1 \exp(-1.1 \times x_1) + w_2 \exp(-1.5 \times x_2) + \varepsilon$

[5.0 marks]

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Q.4. Consider a linear regression model with n data points, where y_i is the actual outcome, $\hat{y}_i$ the model prediction. Which of the following expressions corresponds to a L_2 loss function?

(a) $\sum_{i=1}^{n}(y_i - \hat{y}_i)$

(b) $\sum_{i=1}^{n}|\hat{y}_i - y_i|$

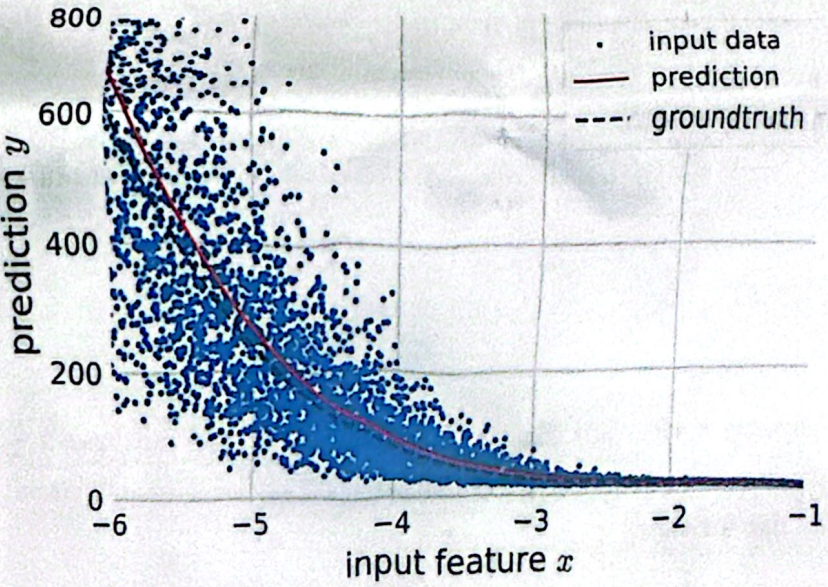
(c) $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

(d) $\sqrt{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$

(e) $\sum_{i=1}^{n} y_i \ln(\hat{y}_i)$

[5.0 marks]

Q.5. We are applying a regression to fit a polynomial to the following dataset. The predicted model does not fit the data very well. Why? (choose one)



(a) there is overfitting

(b) there is underfitting

(c) the training set is too small or not well distributed

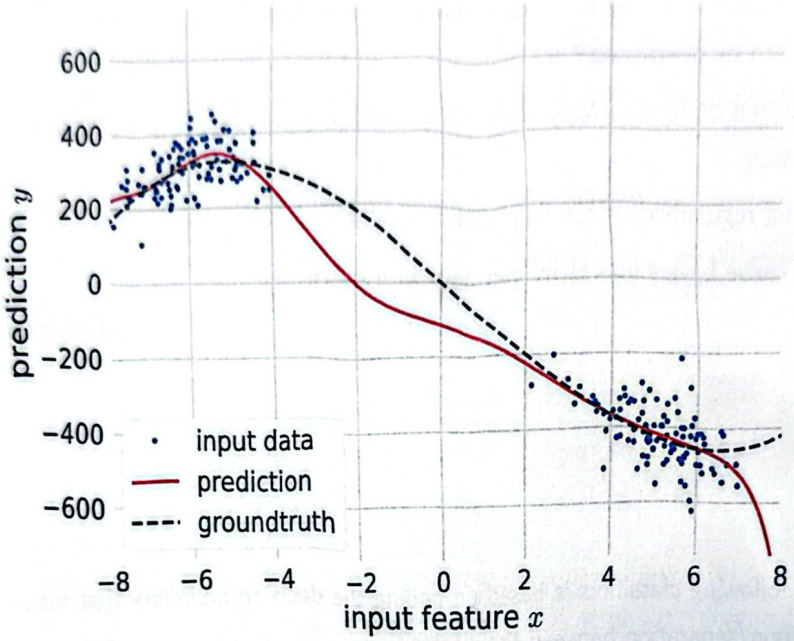
(d) the error is not independent and Gaussian

[5.0 marks]

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Q.6. We are applying a regression to fit a polynomial to the following dataset. The predicted model does not fit the data very well. Why? (choose one)



(a) there is overfitting

(b) there is underfitting

(c) the training set is too small or not well distributed

(d) the error is not independent and Gaussian

[5.0 marks]

Q.7. Large values of the log-likelihood statistic indicate: (choose one)

(a) That there are a greater number of explained vs. unexplained observations.

(b) That the statistical model fits the data well.

(c) That as the predictor variable increases, the likelihood of the outcome occurring decreases.

(d) That the statistical model is a poor fit of the data.

[5.0 marks]

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Q.8. Which of the following statements are correct? (mark all correct answers)

(a) When a model is unable to capture the underlying patterns in the training data, it is said to be overfitting

(b) A model that performs poorly on both the training and validation sets is probably underfitting.

(c) Increasing regularisation can help with underfitting.

(d) Adding Noise Layers in a DNN can help with overfitting.

[5.0 marks]

Q.9. Which of the following classifiers is based on finding the decision boundary that maximizes the margin/separation between the classes?

(a) Logistic Regression

(b) Support Vector Machine (SVM)

(c) Kernel Methods

(d) Decision Tree

(e) k-Nearest Neighbours

[5.0 marks]

Q.10. What is the formula for F1-score?

(a) $F1 = \dfrac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

(b) $F1 = \dfrac{\text{Precision} + \text{Recall}}{2}$

(c) $F1 = \text{Precision} \cdot \text{Recall}$

(d) $F1 = \dfrac{1}{2}\text{Precision} \cdot \text{Recall}$

[5.0 marks]

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Q.11. A Receiver Operating Characteristic (ROC) curve is a graphical plot that illustrates the trade-off between:

(a) True Positive Rate (TPR) and False Positive Rate (FPR)

(b) Precision and Recall

(c) Accuracy and F1-score

(d) Accuracy and Recall

[5.0 marks]

Q.12. Consider a binary classifier with the following confusion matrix:

Confusion Matrix	actual:0	actual:1
predicted: 0	TN=22	FN=4
predicted: 1	FP=8	TP=66

What is the Precision? (one of the proposed solutions is correct)

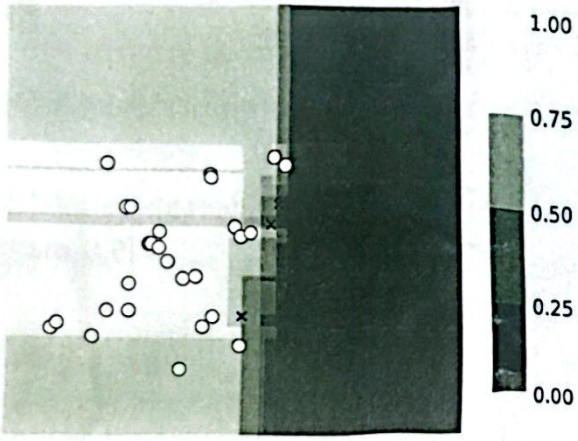
(a) 10.81%
(b) 26.67%
(c) 88.00%
(d) 89.19%
(e) 91.67%
(f) 94.29%

[5.0 marks]

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Q.13. Which of the following binary classifiers produced the following decision boundary map? (choose one)



(a) 5-nearest neighbours
(b) Decision Tree
(c) Adaboost or Random Forest
(d) SVM with RBF kernel
(e) SVM with Polynomial kernel

[5.0 marks]

Q.14. What is the role of the activation function? (choose one)

(a) Calculating the output of the neural network.
(b) Initializing the weights and biases.
(c) Combining multiple inputs values into a single value.
(d) Introducing non-linearity into the network.

[5.0 marks]

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Q.15. What is backpropagation?

(a) A method to update the weights and biases of a DNN.

(b) A method to compute the gradients of the loss function with respect to the model's parameters.

(c) A method to initialise the weights and biases of the neural network.

(d) A propagation of the input values through the network.

[5.0 marks]

Q.16. Consider the network of operations:

$$a = \sin\left(\frac{1}{2} + 4b\right),\ c = b + 2,\ e = a + b + c$$

Which of the following statements is true?

(a) BackPropagation will compute $\frac{\partial e}{\partial a}$ after $\frac{\partial e}{\partial c}$

(b) BackPropagation will compute $\frac{\partial e}{\partial a}$ before $\frac{\partial e}{\partial c}$

(c) BackPropagation can compute $\frac{\partial e}{\partial a}$ and $\frac{\partial e}{\partial c}$ in any order.

(d) none of the above.

[5.0 marks]

Q.17. What is the role of learning rate in optimisation algorithms?

(a) It determines the initial weights and biases of the neural network.

(b) It determines the number of epochs to train the network.

(c) It determines the step size taken in the opposite direction of the gradient.

(d) It determines the direction of the gradient.

[5.0 marks]

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Q.18. What common problem that can cause optimisation to slow convergence or getting stuck in local minima? (choose one)

(a) Overfitting

(b) Using a minibatch that is too small

(c) Vanishing gradients

(d) Using a learning rate that is too high

[5.0 marks]

Q.19. Based on the following code, what is the shape of the output?

```
conv = nn.Conv2d(in_channels=7, out_channels=1, kernel_size=5,
                 stride=3, padding=2)
x = torch.randn(7, 18, 11)
x = conv(x)
```

(a) $1 \times 5 \times 3$

(b) $1 \times 6 \times 4$

(c) $1 \times 7 \times 5$

(d) $1 \times 7 \times 7$

(e) $1 \times 14 \times 7$

(f) $1 \times 16 \times 9$

[5.0 marks]

Q.20. Based on the same code as in previous question, what is the **number of trainable parameters** of the model?

```
conv = nn.Conv2d(in_channels=7, out_channels=1, kernel_size=5,
                 stride=3, padding=2)
x = torch.randn(7, 18, 11)
x = conv(x)
```

(a) 26

(b) 27

(c) 175

(d) 176

(e) 177

(f) 178

[5.0 marks]

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